

Introduction to Differential Equations

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1 Introduction

1.1 Introduction to ODEs

An example of ODE is as follows:

$$\dot{x} = -\alpha x,$$

where $\alpha > 0$. This represents exponential decay, whose solution is $x(t) = e^{-\alpha t}x_0$.

A more complicated example is the logistic equation where $P(t)$ represents the population density of some species, and we have

$$\dot{P} = rP \left(1 - \frac{P}{K}\right),$$

where $r, K > 0$. The solution is

$$P(t) = \frac{P_0 K e^{rt}}{(K - P_0) + P_0 e^{rt}}.$$

Another related example is Lotka-Volterra system. Let $x(t)$ be the population density of the rabbits and $y(t)$ is the population density of foxes. They can be described by the following model:

$$\begin{aligned}\dot{x} &= \alpha x - \beta xy, \\ \dot{y} &= -\gamma y + \delta xy,\end{aligned}$$

where the parameters $\alpha, \beta, \gamma, \delta > 0$.

1.2 Autonomous linear systems

Let $x : [0, \infty) \rightarrow \mathbb{R}^d$ and $A \in \mathbb{R}^{d \times d}$. Consider the equation $\dot{x} = Ax$. The solution to this equation is $x(t) = e^{tA}x_0$, where for any square matrix B we defined

$$e^B = \sum_{n=0}^{\infty} \frac{B^n}{n!}.$$

It is easy to check that this converges nicely, and thus we can take the derivative of e^{tA} against t term by term. We can then verify that $x(t) = e^{tA}x_0$ indeed solves the equation.

There are still something to say about this – how do we compute e^{tA} ? If $UAU^{-1} = D$ for some D diagonal, we then have $e^{At} = U^{-1}e^{Dt}U$. Now if the diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_d)$, we have $e^{Dt} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_d t})$.

However, if A is not diagonal, we can consider the Jordan normal form. Consider a basis change by setting $y = Ux$. We have

$$\dot{y} = U\dot{x} = UAx = UAU^{-1}y.$$

Therefore by choosing matrix U appropriately, we can guarantee UAU^{-1} is in Jordan normal form. Hence now we assume A is in Jordan normal form. Now decompose $\mathbb{R}^d$ into invariant subspaces $\{E_j\}$ with eigenvalues $\lambda_j = a_j + ib_j$. There are two cases.

- If $b_j = 0$, then on E_j , we have

$$(A)_{B_j} = \begin{bmatrix} a_j & 1 & & \\ & a_j & \ddots & \\ & & \ddots & 1 \\ & & & a_j \end{bmatrix}$$

Then,

$$(e^{tA})_{B_j} = e^{ta_j} \sum_{k=0}^{m_j-1} \frac{t^k}{k!} \left((A)_{B_j} - a_j I \right)^k,$$

where $m_j = \dim(E_j)$.

- If $b_j \neq 0$, ***** TO-DO *****